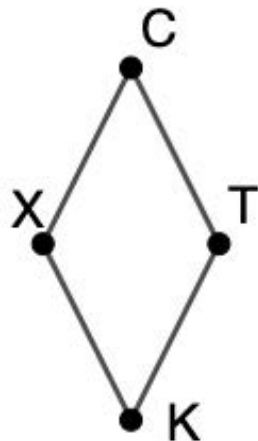


Geometry
Unit 8
Lesson 3 Practice

Name **Answer Key** _____
Date _____
Period _____

Rhombus $CTKX$ is shown where $CT = 15x - 3$ and $TK = 7x + 1$.

1. Select all true properties of $CTKX$.
 - ☐ Opposite sides are parallel.
 - ☐ Opposite sides are congruent.
 - ☐ Consecutive angles are congruent.
 - ☐ All angles are right angles.
 - ☐ Diagonals are perpendicular.
 - ☐ All sides are congruent.



2. What is the value of x ?

$$x = \frac{1}{2}$$

3. What is the length of $\overline{CT}$?

$$CT = 4.5$$

Rectangle $TRXP$ is shown with diagonals $\overline{TX}$ and $\overline{RP}$ which intersect at point Z .

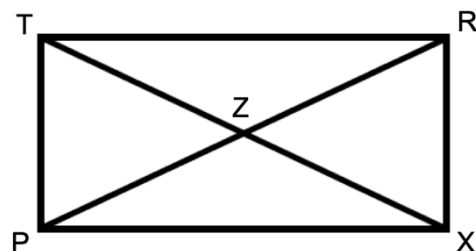
4. Write two congruent statements using the diagonals of $TRXP$.

$$\overline{TX} \cong \overline{RP}$$

$$\overline{TZ} \cong \overline{XZ}$$

$$\overline{PZ} \cong \overline{RZ}$$

$$\overline{TZ} \cong \overline{XZ} \cong \overline{PZ} \cong \overline{RZ}$$



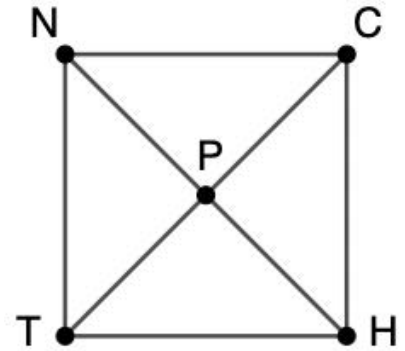
5. $TZ = (5y - 8)$, and $ZX = (3y + 16)$. What is the value of y ?

$$y = 11$$

6. What is the length of $\overline{RP}$?

$$RP = 98$$

Quadrilateral $NCHT$ is shown with diagonals $\overline{NH}$ and $\overline{CT}$ intersecting at point P .



7. What value of x makes $NCHT$ a square, if $m\angle TNC = 90^\circ$, $m\angle NCH = 90^\circ$, $CH = 21$, and $HT = (4x - 7)$?

$$x = 7$$

8. Given $NCHT$ is a square with $PC = (5y - 2)$ and $PT = (3y + 4.74)$. What is the length of $\overline{NH}$?

$$y = 3.37$$

$$NH = 29.7$$

9. What are some similarities between $CTKX$ and $NCHT$?

Opposite sides are parallel

Opposite sides are congruent

Opposite angles are congruent

Consecutive angles are supplementary

Diagonals are perpendicular to each other

Diagonals bisect each other

All sides are congruent

10. What are some differences between $TRXP$ and $NCHT$?

In $NCHT$ diagonals are perpendicular to each other

In $NCHT$ all four sides are congruent